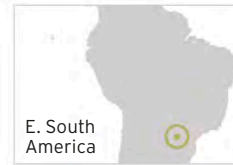


Itaipu Dam

A series of vast structures that together generate 15 percent of the electricity used in Brazil



Spanning the Paraná River between Brazil and Paraguay, the Itaipu Dam is actually a series of four different dams: there are an earthfill dam, a rockfill dam, the concrete main dam, and a concrete wing dam. With a maximum height of 643 ft (196 m), the dams together stretch for 25,981 ft (7,919 m). Behind them is a reservoir with a surface area of 520 square miles (1,350 square km) containing 7 cubic miles (29 cubic km) of water. In 2016, the dam set a world record for producing electricity of 103,098,366 megawatt hours (MWh), surpassing the Three Gorges Dam in China. Half of that electricity goes west to Paraguay, half east to Brazil, although most of the output to Paraguay is subsequently exported back to Brazil.

The sounding stone dam

The dam gets its name from an island in a river nearby. In the local Guaraní language, *itaipu* means “the sounding stone.” The idea for this vast project began in the 1960s, when the neighboring countries agreed to exploit the Paraná River for hydroelectricity. Construction began in January 1971, the river was rerouted in 1978 to allow the dam to be built, and the dam opened for electricity generation on May 5, 1984.



△ GARGANTUAN PROPORTIONS

Around 434 million cubic ft (12.3 million cubic m) of concrete was used to build the four dams—an amount sufficient to build more than 200 football stadiums. The quantity of iron and steel used in the construction could have built 380 Eiffel Towers.

▽ GRAND ILLUMINATION

The illumination of the Itaipu Dam at night highlights the structural and engineering complexity of the massive dam wall.

INSIDE THE DAM

The main dam consists of large concrete segments joined together to enclose a hollow chamber. Across the four dams, water flows through 20 sets of turbines to generate electricity, while excess water from the reservoir not needed for power generation is discharged through 14 separate spillways at a rate of 2,196,572 cubic ft (62,200 cubic m) per second.

